Situated Learning

Multiple Demonstrations

Task: Unzip the bag.

Ep. 1 · Ep. 2 · … · Ep. N

Situation Parsing

Coarse Segmentation

Gripper Events: Open · Close · Open

Clip 1 · Clip 2 · Clip 3 · Clip 4

Foundation Model Reasoning

Semantic phase refinement: Adjust → Grasp → Unzip

Ex. Phase Contract: Grasp

GOAL	Stable grasp	SKILL	Control
PRE.	Pull exposed		…

Phase-wise Skill Construction

Phase-Conditioned VLA Training

Phase-segmented demonstrations: Adjust bag · Unzip bag

Progress Prediction: 65%

Phase-Conditioned Vision-Language-Action Model

Situation-Guided Grasp Control

Demonstration example: Grasp zipper

SAM Segmentation · Path Calculation · Curobo Planning

Learned Grasp Waypoint

Task structure & phase contracts

Prepared skills

SITUATION

Task context — Goal · Object relations

Phase structure — Phases · Dependencies

Phase contracts — Conditions · Recovery

Prepared skills — VLA policies · Guided control

Task structure, contracts & capabilities

Observe & ground

Instruction: Unzip the bag.

Observation

Robot proprioception: EEF pose · Joint positions · Gripper state

Acquire Observations

Situation-guided orchestration

Determine current phase → Check preconditions → Ready → Prepare skill inputs

Unmet

Verification result

Outcome Verification

Situation reference + Current observation → VLM → VLM Check

Ready · Unmet · Uncertain

Skill Execution

Skill Dispatch

Phase-Conditioned VLA: VLA execution → Progress ≥ τ (Yes / No)

Situation-Guided Control: Sequential waypoint execution (w_1, w_2, …, w_n)

RETURN

Execution Result

Situated Agent